From legal research idea to extraction templates: Can AI agents bridge the gap?

Schematize: An Agentic System for Generating and Refining Information-Extraction Schemas for Legal Research

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Example: from a two-sentence idea to a tested schema

1 The idea

HUMAN IN THE LOOP

Legal researcher: "I would like to study **personal-rights violation** cases and assess **how severely** the perpetrator violated them."

Two sentences. That is all the system gets. Scope, law, variables and the real hypothesis are still in the researcher's head.

2 The interview

INTERVIEW AGENT

How do courts weigh severity?

Yes, but mostly what people sue over: are cases getting more trivial year by year?

Severity criteria: reach, intent...?

Above all, what the perpetrator did or said.

2 of 10 real clarifying questions, translated from Polish.

3 First draft

NO RULINGS READ · 5 FIELDS

```
violation_type: enum
[reputation, privacy, image]
severity: int 0--5
court_found_violation: bool
compensation_awarded: bool
compensation_amount: float
```

Drafted from the problem definition, then critiqued: clear? complete? extractable? No agent sees documents yet.

4 Final schema

TESTED ON RULINGS · +5 FIELDS

```
violation_type: enum
severity: int 0--5
court_found_violation: bool
compensation_awarded: bool
compensation_amount: float
+ perpetrator_act: text
+ violation_place: enum
[public, private, workplace]
+ social_media: bool
+ legal_basis: enum
[art.23, art.24 KC, other]
+ judgment_year: int
```

Green: added after reading real rulings. Fields are illustrative.

Rethinking schema evaluation: an expert-question coverage protocol

BENCHMARK DATA

3 legal research problems
7 legal advisers, 3 per problem

- **Drug offences** — 28 questions per expert
- **Medical errors** — 33 questions per expert
- **Personal rights** — 26 questions per expert

One expert's questions	Schema field	LLM judge
Which personal right was violated?	`violation_type`	✓
What compensation was awarded?	`compensation_amount`	✓
Was it published on social media?	`social_media`	✓
What were the circumstances?	— no field	✗

... about 30 questions per expert, written before seeing any schema.

Coverage

Share of an expert's questions that the schema can answer, averaged over each problem's experts.

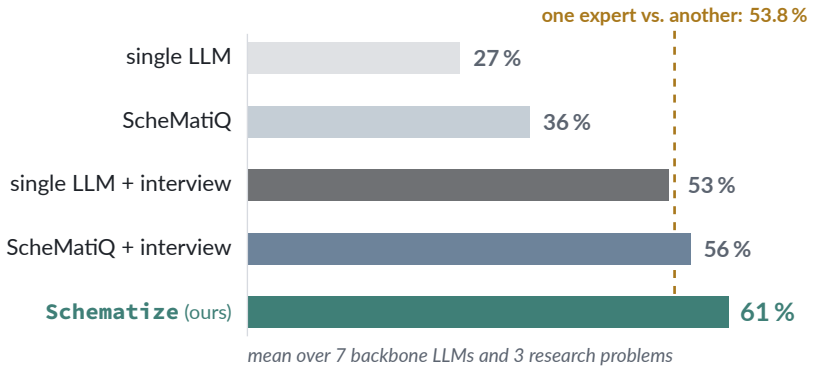
This example: 3 of 4 questions covered.

Schemas from `Schematize` or a baseline. No gold schema needed.

How it works

1. **Interview** HUMAN
   The system asks the researcher what the study should cover, then writes a formal problem definition.
2. **Search queries**
   It writes search queries that will later retrieve court rulings.
3. **Draft without rulings**
   A first schema is drafted and critiqued two or three times before any ruling is read.
4. **Test on rulings**
   Rulings are retrieved; each round reads one and adds missing fields.
5. **Report and chat** HUMAN
   The researcher sees why each field exists and asks for changes in plain language.

Does it work? Mean coverage of expert questions



mean over 7 backbone LLMs and 3 research problems

Takeaways

4 of 7 LLMs beat the expert-vs-expert line.

6 of 7 LLMs: best or 2nd-best.

−14.3 pp without the interview.

$0.02 per problem with Llama 4 Scout.

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`pip install schematize`

open-source, model-agnostic Python library

Code & paper →
github.com/pwr-ai/schematize
pwr-ai.github.io/schematize